

1 Data generation machinery for N-of-1 clinical trial simulations
2 under a two-component response decomposition

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39 1 Abstract

40 **Background.** Simulation studies of aggregated N-of-1 trials must choose where in the joint
41 distribution a biomarker-treatment interaction lives. A companion paper compares two such
42 choices, mean moderation and covariance moderation, under a three-component decomposi-
43 tion of treatment response into pharmacological (BR), placebo-belief (PB), and time-variant
44 natural-history (TV) components. The TV component carries no biomarker coupling in either
45 architecture, so its role in that comparison is not obvious.

46 **Methods.** We repeat the comparison under a reduced data-generating process carrying only
47 BR and PB. Dropping TV reduces the correlation matrix from $(2 + 3n)$ to $(2 + 2n)$, from
48 26×26 to 18×18 at the eight occasions used throughout this program. We cross two
49 architectures against three trial designs, two sample sizes, three biomarker-moderation levels,
50 and three carryover half-lives, holding the analysis model fixed. Appendix C derives the
51 positive-definiteness constraint on the moderation parameter in closed form and characterizes
52 what the reduction does and does not relax.

53 **Results.** *[PLACEHOLDER: pending the production run. To report: the carryover-response*
54 *profile of each architecture under the reduced DGP; Type I error at the null; the comparison*
55 *of absolute power against the three-component setting.]*

56 **Conclusions.** *[PLACEHOLDER: pending results. Expected to state the conditions under*
57 *which the reduction is appropriate for methodological work and the conditions under which the*
58 *full decomposition remains necessary.]*

59 2 1. Introduction

60 An aggregated N-of-1 trial pools repeated within-participant on-drug and off-drug contrasts to
61 estimate a population-level biomarker-treatment interaction¹. Simulation studies of such trials
62 must specify a data-generating process, and that specification involves two choices usually
63 made implicitly. The first is where the interaction lives in the joint distribution of biomarker
64 and response. The second is how many distinct latent components the response trajectory is
65 decomposed into.

66 The companion paper 01 takes up the first choice. It compares *mean moderation*, in which the
67 biomarker shifts the mean of the pharmacological response, against *covariance moderation*, in

68 which the biomarker’s correlation with that response is treatment-state dependent. It shows
69 the two are not interchangeable: they respond differently to carryover, and the difference is
70 large enough to change a design recommendation.

71 The present paper takes up the second choice. The program’s standard DGP decomposes
72 response into three components: BR, the pharmacological action of the compound; PB, the
73 improvement driven by the participant’s belief that treatment will work; and TV, drift in
74 the underlying condition independent of treatment. That decomposition is developed and
75 defended in paper 06, whose subject is the inferential cost of failing to decompose.

76 There is a specific reason to ask whether all three are needed for *methodological* work. The
77 biomarker couples to BR in both architectures. It couples to PB only through an optional
78 contamination parameter that is zero by default. It does not couple to TV at all: the package
79 provides no parameter for a biomarker-TV correlation, so that entry of the correlation matrix
80 is identically zero in every simulation this program has run. Paper 06’s omitted-variable
81 identity then implies that TV cannot displace the biomarker-treatment slope, and paper 06’s
82 Study A confirms this empirically, finding no trend in bias across its TV magnitude axis.

83 If TV cannot bias the estimand, what does it do? It contributes variance. It also contributes
84 n rows and columns to the correlation matrix, two cross-component blocks, and a set of
85 positive-definiteness constraints. Whether that cost is repaid in a study whose object is the
86 comparison between architectures is the question this paper answers, by repeating paper 01’s
87 comparison without it.

88 We hold everything else fixed. The trial designs, the analysis model, the parameter calibration,
89 the biomarker-moderation levels, and the carryover half-lives are those of paper 01. Section 2
90 states the model and the two architectures under the reduced decomposition, Section 3 gives
91 the results, and Section 4 takes up what follows for the design of future simulation work in
92 this program. Three appendices carry the structural material: Appendix A on why the two
93 architectures are treated as a pair, Appendix B on the correlation matrix the reduced DGP
94 assembles, and Appendix C on the positive-definiteness constraint the reduction was expected
95 to relax.

96 **3 2. Methods**

97 **3.1 2.1 Notation and analysis model**

98 Let $i = 1, \dots, N$ index participants and $p = 1, \dots, n$ measurement occasions at cumulative
99 weeks $w_1 < \dots < w_n$. Write B_i for the participant’s baseline biomarker, BL_i for baseline
100 symptom severity, and $S_{x,ip}$ for the observed symptom score. Under the two-component
101 decomposition,

$$102 \quad S_{x,ip} = BL_i - (BR_{ip} + PB_{ip}),$$

103 so that a beneficial response lowers the score. The three-component model of paper 06 adds
104 TV_{ip} inside the parentheses; setting it to zero recovers the present model exactly.

Both components are drawn jointly across occasions from a multivariate normal whose means follow modified Gompertz trajectories,

$$\text{mg}(t; a, d, r) = a \exp(-d \exp(-rt)),$$

evaluated at a component-specific time argument: cumulative time on drug t_{od} for BR, and accumulated belief exposure t_{pb} scaled by the design's expectancy weight η for PB. The distinct time arguments are what make the components separable from design contrasts.

The analysis model is held fixed across all conditions,

$$S_{x,ip} = \beta_0 + \beta_{bm}B_i + \beta_t t_p + \beta_D D_{bc,ip} + \beta_{bm:D} B_i D_{bc,ip} + u_i + \varepsilon_{ip},$$

fitted with `nlme::lme`, a participant random intercept $u_i \sim N(0, \sigma_u^2)$, and continuous-time AR(1) residual correlation `corCAR1(form = ~ t | ptID)`. Here $D_{bc,ip}$ is the continuous exposure indicator, equal to one on drug and decaying as $\exp(-\lambda t_{sd,ip})$ off drug with $\lambda = \ln 2/t_{1/2}$. The estimand is $\beta_{bm:D}$. Appendix A.4 states why the analysis model is held fixed.

3.2 Two data-generating architectures

3.2.1 2.2.1 Mean moderation (Architecture A)

Under mean moderation the biomarker shifts the conditional mean of the pharmacological component at on-drug occasions. After the multivariate normal draw, BR is displaced by

$$BR_{ip} \leftarrow BR_{ip} + c_{bm} z_i \sigma_{BR} \quad \text{for occasion } p \text{ on drug,}$$

where $z_i = (B_i - \mu_{bm})/\sigma_{bm}$ is the standardized biomarker. The covariance structure is untouched: the biomarker is independent of every component draw, and the interaction is carried entirely by the first moment. In the notation of Appendix B, mean moderation sets the coupling vector $b \equiv 0$.

3.2.2 2.2.2 Covariance moderation (Architecture B)

Under covariance moderation the means are untouched and the interaction is carried by a treatment-state-dependent correlation,

$$\text{Cor}(B_i, BR_{ip}) = c_{bm} \cdot \phi(t_{sd,ip}), \quad \phi(t_{sd}) = \begin{cases} 1, & \text{on drug} \\ \exp(-\lambda t_{sd}), & \text{off drug.} \end{cases}$$

The biomarker is more strongly correlated with the pharmacological response while the drug is present, and that association decays after discontinuation at the pharmacokinetic rate.

3.2.3 2.2.3 What the reduced decomposition changes

Neither architecture references TV. Removing it therefore leaves both definitions above unchanged and alters the simulation in three mechanical ways.

First, the correlation matrix shrinks, from $2 + 3n$ to $2 + 2n$. Appendix B gives the resulting block partition.

Second, the outcome loses a variance term. At the reference calibration the three components carry standard deviations of 10 (TV), 10 (PB), and 8 (BR), so TV accounts for approximately 38% of the component variance. Removing it reduces the residual variation against which the interaction is tested, and absolute power rises accordingly. This is an expected arithmetic consequence rather than a finding, and it is why absolute power in this paper is not comparable to paper 01's.

Third, the cross-component structure simplifies. The three-component matrix carries three cross-component blocks; the two-component matrix carries one. Appendix B.6 sets out exactly what is removed, and Appendix C shows that what is removed is not what constrains the moderation parameter.

3.3 2.3 The carryover recursion

Both architectures apply carryover to the BR mean trajectory: at an off-drug occasion, a decayed fraction of the pre-discontinuation response persists. The implementation used here differs from the one behind paper 01's published results, and the difference is worth stating.

The earlier implementation computed

$$\mu_{BR,p} \leftarrow \mu_{BR,p} + \mu_{BR,p-1} (1/2)^{t_{sd,p}/t_{1/2}},$$

in which $\mu_{BR,p-1}$ is the *already adjusted* value from the preceding occasion while $t_{sd,p}$ is *cumulative* time since discontinuation. Elapsed time is therefore counted twice, and every off-drug occasion after the first in an uninterrupted run is over-decayed. At a one-week half-life the second off-drug occasion receives half the residual effect it should.

The present implementation anchors the decay to the mean at discontinuation,

$$\mu_{BR,p} \leftarrow \mu_{BR,p} + \mu_{BR,p^*} (1/2)^{t_{sd,p}/t_{1/2}},$$

where p^* indexes the most recent on-drug occasion, so the cumulative decay factor is applied once. We verified at three reference cells (Hybrid, OL+BDC, and CO at $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$, 400 replicates under a common seed) that the correction moves power by at most 0.005 against a Monte Carlo standard error of about 0.025. It is adopted because it is correct, not because it changes results.

3.4 2.4 Simulation design

The grid crosses architecture, trial design, sample size, biomarker-moderation strength, and carryover half-life.

Factor	Levels
Architecture	mean moderation (A), covariance moderation (B)
Trial design	CO, Hybrid, OL+BDC
Sample size N	35, 70
Biomarker moderation c_{bm}	0, 0.30, 0.45
Carryover half-life $t_{1/2}$	0, 0.5, 1.0 weeks
Components	BR + PB
Replicates	500

This gives $2 \times 3 \times 2 \times 3 \times 3 = 108$ cells. The $c_{bm} = 0$ rows are the Type I error check against nominal 5%; the $t_{1/2} = 0$ rows give a no-carryover reference against which the cost of nonzero carryover is read off directly.

Throughout, N is the **total** number of participants, allocated as evenly as possible across the design's randomization paths, so a design with P paths assigns about N/P to each.

Absolute power is not comparable between architectures at the same nominal c_{bm} , since the parameter is not calibrated to equal power across the two channels. The interpretable comparison is the carryover profile within each architecture, and the qualitative shape of that profile between them.

4 3. Results

[PLACEHOLDER pending the production run.]

Intended structure, mirroring paper 01 Section 3:

- **3.1 Power under carryover at $N = 70$.** Table of power by architecture, design, and $t_{1/2}$ at $c_{bm} = 0.45$, with MCSEs and non-convergence rates. The question is whether the covariance arm's decline with $t_{1/2}$ and the mean arm's flatness, reported by paper 01 under three components, survive the reduction.
- **3.2 Mechanism.** Whether the divergence is attributable to the two channels through which carryover attacks covariance moderation (mean blurring and correlation erosion) against the one channel available under mean moderation.
- **3.3 Type I error and estimator bias.** Rejection rate at $c_{bm} = 0$ against nominal 0.05, across designs, sample sizes, and half-lives.
- **3.4 Comparison with the three-component setting.** Absolute power against paper 01's published figures, with the variance argument of Section 2.2.3 as the expected explanation. Note that this comparison is qualitative: paper 01's figures were produced

191 under the earlier carryover recursion and a partly different grid, so a precise decompo-
192 sition of the difference into its variance and recursion parts is not available without a
193 matched run.

194 5 4. Discussion

195 *[PLACEHOLDER pending results.]*

196 Intended structure:

- 197 • **4.1 What the TV component is for.** The narrow claim the results can support
198 (TV is inert for this estimand under the DGP as implemented) against the broad claim
199 they cannot (that TV is dispensable). Three surviving roles: attribution estimands,
200 biomarkers correlated with natural-history drift, and absolute power figures calibrated
201 against a clinical reference.
- 202 • **4.2 When the reduction is appropriate.** Proposed division: reduced DGP for
203 comparative methodological work, full DGP whenever a number is read as an estimate
204 of what a real trial would achieve or the decomposition is itself the object of study.
- 205 • **4.3 A gap this paper exposes.** Paper 06 states its identity over both non-
206 pharmacological components and recommends decomposition when a biomarker
207 may correlate with PB *or* TV, but the package implements `c.bm.pb` and no TV
208 counterpart. The TV half of that claim has never been exercised. Our own conclusion
209 is correspondingly conditional: TV is inert under the DGP as implemented, not
210 intrinsically.
- 211 • **4.4 Limitations.** Single parameter calibration; removal rather than flattening (Section
212 2.2.3 and Appendix B.6); no matched three-component run.

213 6 5. Conclusions

214 *[PLACEHOLDER pending results.]*

215 7 Conflict of interest

216 The authors declare no conflicts of interest.

217 8 Funding

218 This work received no specific funding.

219 9 Reproducibility

220 The simulation driver is `analysis/scripts/two-component-dgp/01-run-grid.R` and its
221 output is versioned at `analysis/data/13-two-component-grid.rds`. The driver sets a mas-

222 ter seed and derives a per-cell seed from it. Every number quoted in the results is read from
223 that file at render time rather than typed in.

224 The two-component decomposition is supported by the `components` argument added to
225 `generateData()` and `buildSigma()` in `R/generateData.R`. It defaults to `c("tv", "pb",`
226 `"br")`, so existing calls are unaffected; passing `c("pb", "br")` produces the reduced
227 matrix used here. The argument requires that `"br"` be retained, since BR carries the
228 interaction channel in both architectures, and canonicalizes the ordering so that label and
229 standard-deviation sequence is stable regardless of how the caller writes it. The package test
230 suite passes unchanged under the default.

231 The corrected carryover recursion of Section 2.3 is applied in `R/generateData.R` and
232 `implementations/tidyverse/R/functions.R`. Other implementation collections retain the
233 earlier form; see `02-deviations.md` in the driver directory.

234

235 10 Appendix A. Why mean and covariance moderation

236 10.1 A.1 The two channels exhaust the Gaussian case

237 The simulations draw participant trajectories from a multivariate normal distribution, and a
238 multivariate normal is completely determined by its mean vector μ and its covariance matrix
239 Σ . Any biomarker-treatment interaction written into such a data-generating process must
240 therefore enter through μ , through Σ , or through both. Mean moderation is the first route,
241 covariance moderation the second, and the combined architecture developed in a companion
242 paper² is the third. Within the Gaussian family there is no fourth place to put the interaction.
243 This is why the two architectures are treated as a pair rather than as two entries in an open-
244 ended list: they are not two options among many, but the two components of a decomposition.

245 Reducing the response decomposition from three components to two does not disturb this
246 argument. The exhaustiveness claim concerns where an interaction can be written into a
247 Gaussian joint distribution, not how many latent components that distribution is built from.
248 Dropping TV removes rows from Σ ; it does not create a third place for the interaction to live,
249 nor remove either of the two that exist.

250 The exhaustiveness claim is specific to the Gaussian family and should not be read more
251 broadly. A data-generating process built on a finite mixture, or on explicit latent responder
252 classes, can carry an interaction in features that neither μ nor Σ captures for a single compo-
253 nent, and Senn notes finite mixtures as an alternative variance-component model³. Covariance
254 moderation is best understood as the second-moment reduced form of such a mixture, which
255 is why the latent-class treatment is developed separately⁴.

256 10.2 A.2 What each channel asserts

257 The two channels are not merely different parameterizations of one mechanism; each commits
258 to a distinct claim about what the biomarker is.

259 Mean moderation asserts that the biomarker fixes a participant's drug effect. Two participants
260 sharing a biomarker value share an expected treatment effect exactly, and any difference
261 between their realized responses is measurement error. This is the appropriate commitment
262 when the biomarker measures pathway engagement directly, or stands in a causal path between
263 treatment and response.

264 Covariance moderation asserts that the biomarker predicts response only probabilistically.
265 Two participants sharing a biomarker value may still respond differently, because the
266 biomarker indexes an unobserved responder status imperfectly rather than determining
267 response. This is the appropriate commitment when the biomarker is a statistical proxy,
268 which is the situation the prazosin and PTSD application was designed around.

269 The distinction is substantive rather than technical. It determines what a fitted interaction
270 coefficient means, and how much power carryover costs.

271 10.3 A.3 Standing in the literature

272 Mean moderation is the prevailing convention wherever biomarker-treatment interactions are
273 simulated. Haller and Ulm⁵ place the interaction in a regression mean, as do the oncology
274 design literatures on enrichment and predictive-marker trials^{6,7}. Grenet and colleagues⁸ use
275 the same form for a crossover. That literature is, however, overwhelmingly parallel-group.

276 The N-of-1 methodological literature does not adopt mean moderation, because it does not
277 model an observed moderator at all. Its standard formulation carries a random participant-
278 by-treatment interaction, a variance component describing unexplained heterogeneity with
279 no covariate attached^{3,9,10}. Biomarker-validation simulation for N-of-1 designs is therefore
280 close to unoccupied territory rather than a settled field with a convention to follow, and
281 Hendrickson et al.¹ is its principal entry. The two-channel decomposition of A.1 is adopted
282 precisely because no established N-of-1 convention exists to inherit.

283 10.4 A.4 Holding the analysis model fixed

284 Both architectures are analyzed with the identical model of Section 2.1. This is deliberate
285 and worth stating explicitly, because a data-generating process and an analysis model are
286 separate objects, and a comparison that varies both attributes differences to neither. The
287 continuous exposure variable Dbc loses contrast variance as carryover grows, and a binary on-
288 drug indicator does not. A comparison in which one architecture carried a continuous regressor
289 and the other a binary one would confound that effect with the architecture contrast it was
290 meant to isolate. Holding the analysis model fixed means every difference reported in Section
291 3 is attributable to the data-generating process alone.

292 The same argument governs the reduction studied here. The analysis model does not reference
293 the component decomposition at all: it sees only S_x , and S_x is a sum. Removing TV therefore

changes what the analysis is fitted to without changing what it fits, which is the property that makes the comparison in Section 3 interpretable.

296

11 Appendix B. Correlation structure under the two-component decomposition

This appendix sets out the correlation matrix the reduced data-generating process assembles, in the notation of paper 01's Appendix B so that the two can be read side by side. Symbols follow the variable names used in the code: BM is the biomarker, written B_i in Section 2.1, and BR_p , PB_p the response components at occasion p .

Mean moderation appears below only through the statement that its coupling vector is zero. It writes no biomarker-response correlation into the covariance at all; its interaction is carried in the mean instead, as Section 2.2.1 sets out.

11.1 B.1 Variable ordering and block partition

Each participant's trajectory is drawn from a single multivariate normal over an ordered variable list. With n measurement occasions at cumulative weeks $w_1 < \dots < w_n$, the two-component vector is

$$X = (\text{BM}, \text{BL}, PB_1, \dots, PB_n, BR_1, \dots, BR_n)^\top, \quad \dim X = 2 + 2n,$$

against $2 + 3n$ under the three-component decomposition. The ordering is component-major: all n occasions of one component are contiguous, rather than all components at one occasion. Writing A_c for the $n \times n$ within-component block of component $c \in \{pb, br\}$ and $C_{pb,br}$ for the single cross-component block, the correlation matrix partitions as

$$R = \begin{pmatrix} 1 & 0 & 0^\top & b^\top \\ 0 & 1 & 0^\top & 0^\top \\ 0 & 0 & A_{pb} & C_{pb,br} \\ b & 0 & C_{br,pb} & A_{br} \end{pmatrix}.$$

Three features carry over unchanged from the three-component case. The baseline term BL is uncorrelated with every other variable, so its row and column are those of the identity. The biomarker BM couples only to the response component BR , through the vector $b \in \mathbb{R}^n$; its correlation with PB is zero throughout this paper. And b is the only place in R where the architectures differ.

The third feature deserves emphasis because it is what licenses the reduction. In the three-component matrix the biomarker's row is $(1, 0, 0^\top, 0^\top, b^\top)$: zero against both TV and PB .

Deleting the TV rows and columns therefore deletes only zeros from that row. The coupling vector is untouched, entry for entry.

11.2 B.2 Within-component blocks A_c

Let ρ_c denote the autocorrelation parameter of component c (`c.pb`, `c.br` in the code). The correlation decays with the elapsed time between occasions, using the cumulative week gap rather than the index difference:

$$[A_c]_{ij} = \rho_c^{|w_i - w_j|}.$$

For $n = 4$, writing $d_{ij} = |w_i - w_j|$,

$$A_c = \begin{pmatrix} 1 & \rho_c^{d_{12}} & \rho_c^{d_{13}} & \rho_c^{d_{14}} \\ \rho_c^{d_{12}} & 1 & \rho_c^{d_{23}} & \rho_c^{d_{24}} \\ \rho_c^{d_{13}} & \rho_c^{d_{23}} & 1 & \rho_c^{d_{34}} \\ \rho_c^{d_{14}} & \rho_c^{d_{24}} & \rho_c^{d_{34}} & 1 \end{pmatrix}.$$

Because the exponent uses calendar time, unequally spaced designs produce unequal decay across consecutive occasions. In the Hybrid design, whose occasions fall at weeks 4, 8, 9, 10, 11, 12, 16, 20, consecutive gaps of one week during the discontinuation phase yield much higher adjacent correlations than the four-week gaps at either end.

We note that the code uses the cumulative week gap, not the occasion index difference, and that this choice is consequential: Appendix C.4 reports that the index-based form would cost roughly a fifth of the feasible range of c_{bm} .

11.3 B.3 The cross-component block $C_{pb,br}$

Let c_1 denote the same-occasion cross-component correlation (`c.cf1t`) and $c_\times$ the different-occasion cross-component correlation (`c.cfct`). The same-occasion entry is a constant and the different-occasion entries inherit the AR(1) decay:

$$[C_{pb,br}]_{ij} = \begin{cases} c_1, & i = j \\ c_\times \rho^{|w_i - w_j|}, & i \neq j. \end{cases}$$

Under the three-component decomposition there are three such blocks, $C_{tv,pb}$, $C_{tv,br}$ and $C_{pb,br}$, all of this form. The reduction retains the last and deletes the first two.

11.4 B.4 Biomarker coupling vector b

The vector b carries the biomarker-treatment interaction under covariance moderation. Its entry is gated on the drug indicator and decays with time since discontinuation $t_{sd,p}$ at rate $\lambda = \ln 2/t_{1/2}$:

$$b_p = \begin{cases} c_{bm}, & \text{occasion } p \text{ on drug} \\ c_{bm} e^{-\lambda t_{sd,p}}, & \text{occasion } p \text{ off drug.} \end{cases}$$

Under mean moderation $b \equiv 0$ and the interaction is carried by the displacement of μ_{BR} described in Section 2.2.1. The two architectures are therefore distinguished, in the language of this appendix, by exactly one object: whether b is populated or zero.

11.5 B.5 A worked example

To make the blocks concrete, take four occasions drawn from the Hybrid design at cumulative weeks $w = (10, 11, 12, 16)$, with the first and fourth on drug and the second and third off it, so that $t_{sd} = (0, 1, 2, 0)$. Set $\rho_{br} = 0.6$, $\rho_{pb} = 0.4$, $c_1 = 0.3$, $c_{\times} = 0.1$, $c_{bm} = 0.45$, and a carryover half-life of one week, giving $\lambda = \ln 2 \approx 0.693$. The matrix of week gaps is

$$(|w_i - w_j|) = \begin{pmatrix} 0 & 1 & 2 & 6 \\ 1 & 0 & 1 & 5 \\ 2 & 1 & 0 & 4 \\ 6 & 5 & 4 & 0 \end{pmatrix}.$$

The unequal spacing matters: occasions one to three are a week apart, while the fourth sits four weeks after the third.

11.5.1 B.5.1 Within-component block A_{br}

With $\rho_{br} = 0.6$,

$$A_{br} = \begin{pmatrix} 1 & 0.6000 & 0.3600 & 0.0467 \\ 0.6000 & 1 & 0.6000 & 0.0778 \\ 0.3600 & 0.6000 & 1 & 0.1296 \\ 0.0467 & 0.0778 & 0.1296 & 1 \end{pmatrix}.$$

The fourth occasion is nearly uncorrelated with the first three, a direct consequence of the four-week gap: $0.6^4 = 0.1296$ against $0.6^1 = 0.6$ for an adjacent pair.

11.5.2 B.5.2 Cross-component block $C_{pb,br}$

With $c_1 = 0.3$, $c_{\times} = 0.1$ and the decay taken at $\rho = 0.6$,

$$C_{pb,br} = \begin{pmatrix} 0.3000 & 0.0600 & 0.0360 & 0.0047 \\ 0.0600 & 0.3000 & 0.0600 & 0.0078 \\ 0.0360 & 0.0600 & 0.3000 & 0.0130 \\ 0.0047 & 0.0078 & 0.0130 & 0.3000 \end{pmatrix}.$$

370 The block is symmetric here because the two components share a decay rate in this example;
371 in general it need not be.

372 11.5.3 B.5.3 Biomarker vector b

373 Occasions one and four are on drug, occasions two and three off it with $t_{sd} = 1$ and 2. With
374 $c_{bm} = 0.45$ and $\lambda = \ln 2$,

$$375 \quad b = (0.4500, 0.2250, 0.1125, 0.4500)^\top.$$

376 The off-drug entries halve with each additional week since discontinuation, and the fourth
377 occasion returns to full strength because the participant is back on drug. It is this alternation
378 that carries the interaction, and its erosion under carryover that Section 3 measures.

379 11.6 B.6 What the reduction removes, and what it does not

380 Relative to the three-component matrix, the reduction removes:

- 381 • one within-component block, A_{tv} ($n \times n$);
- 382 • two cross-component blocks, $C_{tv,pb}$ and $C_{tv,br}$;
- 383 • n rows and columns from R , taking it from $2 + 3n$ to $2 + 2n$: at $n = 8$, from 26×26 to
384 18×18 .

385 It removes no entry of b , because the biomarker never coupled to TV . It removes no parameter
386 from the architecture definitions of Section 2.2, because neither references TV .

387 One distinction is worth recording, because it is easy to conflate two operations that are not
388 equivalent. Removing the component is not the same as flattening its trajectory. The code
389 sets component standard deviations independently of the Gompertz maximum, so setting that
390 maximum to zero yields a flat TV mean while leaving the TV variance contribution, its block
391 A_{tv} , and its cross-component blocks in place. A study wanting the simpler mental picture
392 without the change in residual variance should flatten; a study wanting the smaller matrix
393 should remove. This paper removes.

394

395 12 Appendix C. Positive definiteness and the feasible range of

396 c_{bm}

397 The correlation matrix of Appendix B must be positive definite for the multivariate normal
398 draw to be valid. It is not automatically so: the moderation parameter c_{bm} is bounded above,
399 and the bound depends on the correlation structure, the trial design, and the number of

occasions. This appendix derives the bound, gives a closed form for the compound-symmetric case, and reports where the reduction of Appendix B does and does not relax it.

The motivation is practical. Simulation drivers in this program repair a non-positive-definite matrix silently, by projection onto the nearest positive-definite matrix, so a grid cell that violates the bound does not fail. It quietly becomes a different data-generating process, with a realized biomarker-response correlation no longer equal to the nominal c_{bm} . Knowing the bound in advance is therefore a precondition for trusting a grid.

12.1 C.1 The constraint is a Schur complement on the biomarker row

Partition the correlation matrix of Appendix B.1 by separating the biomarker from everything else. Because BL is uncorrelated with all other variables, it contributes an identity block that factors out of every determinant and may be dropped. What remains is

$$R' = \begin{pmatrix} 1 & \tilde{b}^\top \\ \tilde{b} & M \end{pmatrix}, \quad M = \begin{pmatrix} A_{pb} & C_{pb,br} \\ C_{br,pb} & A_{br} \end{pmatrix},$$

where M is the $Kn \times Kn$ component block ($K = 2$ here, $K = 3$ under the full decomposition) and $\tilde{b} \in \mathbb{R}^{Kn}$ is the biomarker coupling extended by zeros: it equals b on the BR positions and vanishes elsewhere.

By the Schur complement, R' is positive definite if and only if M is positive definite and

$$1 - \tilde{b}^\top M^{-1} \tilde{b} > 0.$$

Writing $\tilde{b} = c_{bm} v$, where v is the unit-scaled coupling pattern with $v_p = \phi(t_{sd,p})$ on the BR positions and zero elsewhere, the condition becomes $c_{bm}^2 v^\top M^{-1} v < 1$, giving the exact ceiling

$$c_{bm}^* = (v^\top M^{-1} v)^{-1/2}. \tag{1}$$

Three consequences follow immediately, and they organize the rest of this appendix. The ceiling depends on the component block only through the quadratic form $v^\top M^{-1} v$, so components that are weakly coupled to BR contribute little to it. It depends on the design through v , which encodes the on-drug and off-drug pattern. And it is unaffected by mean moderation, under which $b \equiv 0$ and the constraint reduces to positive definiteness of M alone.

12.2 C.2 Closed form for the component block under compound symmetry

When the within-component blocks are compound symmetric rather than AR(1), M admits an exact spectral decomposition. This is the structure of the published implementation discussed in paper 01, and it is worth recording because it explains a result of C.3 that is otherwise surprising.

Under compound symmetry with a common within-component correlation ρ , common same-occasion cross-component correlation c_1 , and common different-occasion cross-component correlation $c_\times$, the component block is a Kronecker sum,

$$M = A \otimes I_n + B \otimes (J_n - I_n), \quad (2)$$

where J_n is the $n \times n$ matrix of ones and A, B are $K \times K$:

$$A = I_K + c_1(J_K - I_K), \quad B = \rho I_K + c_\times(J_K - I_K).$$

A describes the same-occasion structure across components, B the across-occasion structure. Since $J_n - I_n$ has eigenvalues $n - 1$ once and -1 with multiplicity $n - 1$, and I_n commutes with it, the Kn eigenvalues of M collapse to those of two $K \times K$ reductions:

$$\text{eig}(M) = \underbrace{\text{eig}(A + (n - 1)B)}_{\text{once each}} \cup \underbrace{\text{eig}(A - B)}_{\text{multiplicity } n-1}.$$

Both reductions are of the form $\alpha I_K + \beta(J_K - I_K)$, whose eigenvalues are $\alpha + (K - 1)\beta$ once and $\alpha - \beta$ with multiplicity $K - 1$. Substituting gives all four distinct eigenvalues in closed form:

$$\begin{aligned} \text{all-ones time direction: } & [1 + (n - 1)\rho] + (K - 1)[c_1 + (n - 1)c_\times], \\ & [1 + (n - 1)\rho] - [c_1 + (n - 1)c_\times] \quad (K - 1 \text{ times}), \\ \text{time-contrast directions: } & (1 - \rho) + (K - 1)(c_1 - c_\times), \\ & (1 - \rho) - (c_1 - c_\times) \quad (K - 1 \text{ times, each with multiplicity } n - 1). \end{aligned}$$

At the program's reference values $\rho = 0.7$, $c_1 = 0.2$, $c_\times = 0.1$, $n = 8$, the three-component block has exactly four distinct eigenvalues, 7.7, 5.0, 0.5 and 0.2, the last with multiplicity $2(n - 1) = 14$.

12.3 C.3 Why the component count barely matters

The smallest eigenvalue in C.2 is generally the last,

$$\lambda_{\min}(M) = (1 - \rho) - (c_1 - c_\times), \quad (3)$$

and the striking feature of (3) is what does not appear in it. It contains neither n nor K . Under compound symmetry the positive-definiteness of the component block is independent of both the number of occasions and the number of components. Adding a third component does not make M harder to keep positive definite; nor does adding occasions.

This is the analytic explanation for a result that would otherwise be puzzling. One might expect that deleting a third of the rows and columns from a correlation matrix would materially enlarge the feasible parameter space. It does not, and (3) shows why: the binding structural constraint never depended on the count in the first place.

Two further readings of (3) are worth recording. The operative cross-component quantity is the *difference* $c_1 - c_\times$, not either coefficient separately, so keeping the same-occasion and across-occasion couplings close to one another matters more than keeping either small. And positive definiteness requires $\rho < 1 - (c_1 - c_\times)$, which at the reference values is $\rho < 0.9$: a compound-symmetric structure cannot represent very high autocorrelation at all.

Under AR(1) the corresponding statement is different, and no equally compact closed form is available, because A_c is Toeplitz rather than exchangeable and does not commute with the cross-component structure in the way (2) requires. Numerically, the AR(1) component block becomes progressively more ill-conditioned as occasions are added, since each new occasion is strongly correlated with its neighbours. The number of *occasions* therefore does constrain c_{bm} under AR(1), while the number of *components* still does not.

12.4 C.4 Numerical ceilings

Evaluating (1) directly at the program's reference parameters ($\rho = 0.7$, $c_1 = 0.2$, $c_\times = 0.1$, $n = 8$, AR(1) in cumulative weeks), minimized across each design's randomization paths:

Design	$t_{1/2}$	Three components	Two components
CO	0.0	0.619	0.627
CO	0.5	0.619	0.627
CO	1.0	0.619	0.627
Hybrid	0.0	0.456	0.469
Hybrid	0.5	0.506	0.516
Hybrid	1.0	0.544	0.552
OL+BDC	0.0	0.452	0.464
OL+BDC	0.5	0.500	0.509
OL+BDC	1.0	0.533	0.540

The reduction buys between 0.008 and 0.013 of feasible range. This is consistent with C.3: the deleted blocks were coupled to *BR* only through c_1 and $c_\times$, both small, so they contributed little to the quadratic form $v^\top M^{-1}v$ that sets the ceiling.

Three further observations follow from evaluating (1) under variations of the structure, none of which is specific to the reduction.

First, the ceiling rises with the carryover half-life within the two non-crossover designs, from 0.456 to 0.544 under Hybrid. A longer half-life makes the coupling vector v smoother, and a smoother v aligns better with the well-conditioned directions of M . The crossover design is insensitive to $t_{1/2}$ because its off-drug occasions are spaced at least 2.5 weeks apart, so $\phi(t_{sd})$ has decayed to near zero at every one of them regardless.

Second, the number of occasions matters considerably more than the number of components. Holding the design shape fixed and varying n alone, the ceiling falls from approximately 0.81 at $n = 4$ to 0.58 at $n = 12$ under a three-component structure. That is an order of magnitude larger than the effect of deleting an entire component.

Third, the exponent scale in A_c is consequential. Using the occasion-index difference $|i - j|$ in place of the cumulative week gap $|w_i - w_j|$, an otherwise identical AR(1) structure, lowers the worst-cell ceiling from approximately 0.45 to 0.35. The code uses the week gap; documentation elsewhere in this project has described it as the index difference, and the two are not interchangeable.

12.5 C.5 Practical implications

For the present paper, (1) confirms that the grid of Section 2.4 is feasible throughout: the largest moderation level used is $c_{bm} = 0.45$, against a worst-cell ceiling of 0.464 under the two-component structure. The margin is real but not generous, and a study wishing to extend the grid past $c_{bm} = 0.45$ should evaluate (1) first rather than rely on the simulation to complain, since it will not.

More generally, the appendix identifies which levers move the feasible range and which do not. The number of latent components does not, and neither does the choice between two and three of them. The number of measurement occasions does, substantially. The within-component correlation ρ does, non-monotonically, with the reference value 0.7 close to the maximum of the ceiling. And the cross-component coefficients matter through their difference rather than their magnitudes.

12.5.0.1 Provenance. The Kronecker identity (2) and the eigenvalue decomposition of C.2 were verified numerically against direct construction for $K \in \{2, 3\}$ and $n \in \{4, 8, 12\}$. The ceilings in C.4 were computed from (1) and agree with an independent grid search over c_{bm} in steps of 0.01 to within the grid step. The observations in C.4 concerning occasion count, exponent scale, and ρ were obtained by the same grid search under a reimplemented matrix builder rather than the production `buildSigma`, and should be read as characterizing the structure rather than as production audit results.

13 References

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